

# Human body heat balance equation to consider core body temperature in assessment of heatstroke risk

Tomonori Sakoi<sup>a,\*</sup>, Yoshihito Kurazumi<sup>b</sup>, Sri Rahma Apriliyanthi<sup>c</sup>, Shin-ichi Sawada<sup>d</sup>, Chuansi Gao<sup>e</sup>

<sup>a</sup> Institute of Textile Science and Technology, Academic Assembly, Shinshu University, Ueda, 386-8567, Japan

<sup>b</sup> School of Life Studies, Sugiyama Jogakuen University, Nagoya, 464-8662, Japan

<sup>c</sup> Graduate School of Medicine, Science and Technology, Shinshu University, Ueda, 386-8567, Japan

<sup>d</sup> Faculty of Education, Tokyo University of Social Welfare, Isesaki, 372-0831, Japan

<sup>e</sup> Aerosol and Climate Laboratory, Division of Ergonomics and Aerosol Technology, Department of Design Sciences, Faculty of Engineering, Lund University, Solvegatan 26, 223 62, Lund, Sweden

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## ABSTRACT

Increased core body temperature and dehydration are critical thermal physiological responses to heat disorders. Dehydration is preventable through adequate water intake. However, a simple heat balance equation for assessing the heatstroke risk considering the core body temperature is currently lacking. Hence, this study proposes an equation for evaluating heatstroke risk. To derive the equation, the thermal regulation of the human body and heat interactions between the body core, skin, clothing, and indoor and outdoor environments were considered. The heat flow and heat capacity were represented as water flows caused by the water head difference and cross-sectional area of the tanks, respectively. Subsequently, the importance of considering the above elements as a unified system for designing a thermally comfortable environment and preventing heatstroke is discussed. Next, a heatstroke threshold equation is presented in the form of human body heat balance equation, which considers the core body temperature. This equation enabled us to evaluate the threshold for the combination of the environmental, clothing, and activity parameters beyond which the body temperature exceeds the acceptable limit for heatstroke. The equation was validated by comparing the thresholds with the reported upper limit of the prescriptive zone. The calculated thresholds for various parameters revealed that the parameters related to evaporative heat loss, such as clothing evaporative resistance, were the dominant parameters influencing the threshold. Although the equation can evaluate the influences of various types of activity and clothing, it can be applied only to acclimatized workers in good health with sufficient water and salt intake.

## 1. Introduction

Thermal comfort and heat disorders result from heat interactions between the core of the human body, skin, clothing, and surrounding thermal environment through heat and moisture transportation. Because the core body temperature is regulated within a narrow range depending on the intensity of the activity for a wide range of environmental operative temperatures [1], it is not a sensitive thermal physiological variable for the assessment of thermal comfort. Most heat in the core body moves to the skin before dissipating into the surrounding environment. Therefore, changing the heat transfer characteristics between the core body and skin is essential to maintain a relatively stable

core body temperature when the skin temperature varies widely. Because the skin is exposed to the surrounding environment, the skin temperature and sweat rate change depending on the surrounding environment. Fanger regarded the skin temperature and sweat rate as critical parameters for assessing thermal comfort [2].

Regarding the risk of heat disorders, a shortage of sodium causes neurotransmitter disorders and results in heat cramps [3]. Sweat is made from blood, and the blood volume in the body is limited; hence, the loss of blood volume through sweating reduces the ability to sweat and causes dehydration and heat exhaustion [3]. The recovery of blood volume and sodium concentration through water and sodium intake alleviates heat cramps, dehydration, and exhaustion. Without adequate treatment for heat cramps and exhaustion, the thermal regulation ability

\* Corresponding author. Tokida 3-15-1, Ueda, 386-8567, Japan.

E-mail address: [t-sakoi@shinshu-u.ac.jp](mailto:t-sakoi@shinshu-u.ac.jp) (T. Sakoi).

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| Abbreviations       |                                                                                                |                    |                                                                                                                     |
|---------------------|------------------------------------------------------------------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------|
| ACGIH               | American conference of governmental industrial hygienists                                      | LR                 | Lewis ratio (K/kPa)                                                                                                 |
| CAV                 | clothing adjustment value                                                                      | $M$                | Metabolic rate (W/m <sup>2</sup> )                                                                                  |
| CC                  | cotton coveralls                                                                               | $P_a$              | Environmental vapor pressure (kPa)                                                                                  |
| DLE                 | duration of limited exposure                                                                   | $P_{sk}$           | Vapor pressure at skin (kPa)                                                                                        |
| eWBGT               | equivalent wet-bulb globe temperature                                                          | $P_{sk}^*$         | Saturated vapor pressure at $T_{sk}$ (kPa)                                                                          |
| FLP                 | film-laminated protective clothing                                                             | $Q$                | Total steady heat loss from the core body having $T_{cr, lim}$ , $R_{cs, min}$ , and $SW_{max}$ (W/m <sup>2</sup> ) |
| HSDA                | Heat Strain Decision Aid                                                                       | $R_{e,cl, st}$     | Static evaporative heat transfer resistance of clothing (m <sup>2</sup> kPa/W)                                      |
| HSI                 | heat stress index                                                                              | $R_{cs}$           | Thermal resistance between the body core and skin (m <sup>2</sup> °C/W)                                             |
| ISO                 | International organization for standardization                                                 | $R_{cs, min}$      | $R_{cs}$ at the maximum blood flow (m <sup>2</sup> °C/W)                                                            |
| PHS                 | predicted heat strain                                                                          | $R_{e, T, st}$     | Static clothing total evaporative resistance (m <sup>2</sup> kPa/W)                                                 |
| PMV                 | predicted mean vote                                                                            | $S$                | Heat stored in the human body core under maximum thermal regulation (W/m <sup>2</sup> )                             |
| RH                  | relative humidity                                                                              | $SW_{max}$         | Maximum sweat rate at a given activity, expressed as the latent heat per unit body surface area (W/m <sup>2</sup> ) |
| SWreq               | Required sweat rate                                                                            | $T_a$              | Air temperature (°C)                                                                                                |
| TYC                 | Tychem QC                                                                                      | $T_{cl}$           | Clothing outer surface temperature (°C)                                                                             |
| VPAIP               | vapor-permeable but air-impermeable protective clothing                                        | $T_{cr}$           | Core body temperature (°C)                                                                                          |
| WBGT                | wet-bulb globe temperature                                                                     | $T_{cr, lim}$      | Acceptable upper limit $T_{cr}$ (°C)                                                                                |
| WC                  | work clothes                                                                                   | $T_g$              | Globe temperature (°C)                                                                                              |
| <b>Nomenclature</b> |                                                                                                | $T_{nw}$           | Natural wet-bulb temperature (°C)                                                                                   |
| $C_{res}$           | Sensible heat loss through respiration (W/m <sup>2</sup> )                                     | $T_{op}$           | Indoor operative temperature (°C)                                                                                   |
| $E_{res}$           | Latent heat loss through respiration (W/m <sup>2</sup> )                                       | $T_{out}$          | Outdoor operative temperature (°C)                                                                                  |
| $E_{sk}$            | Evaporative heat loss (W/m <sup>2</sup> )                                                      | $T_{op, lim}$      | $T_{op}$ of the threshold (°C)                                                                                      |
| $E_{max}$           | Evaporation capacity of the surrounding environment including the clothing (W/m <sup>2</sup> ) | $T_r$              | Mean radiant temperature (°C)                                                                                       |
| $f_{cl}$            | Clothing area factor (N.D.)                                                                    | $T_{sk}$           | Skin temperature (°C)                                                                                               |
| $F_{pcl}$           | Vapor permeation efficiency (N.D.)                                                             | $v$                | Air velocity (m/s)                                                                                                  |
| $F_r$               | Radiation area factor of the human body (N.D.)                                                 | $w$                | Skin wettedness (N.D.)                                                                                              |
| $h_c$               | Convective heat transfer coefficient (W/(m <sup>2</sup> °C))                                   | $\alpha$           | Differential coefficient of Antoine equation by $T_{sk}$ (kPa/°C)                                                   |
| $h_r$               | Radiative heat transfer coefficient (W/(m <sup>2</sup> °C))                                    | $\beta$            | Intercept of the linear equation around $T_{sk}$ for the Antoine equation (kPa)                                     |
| $i_{cl}$            | Clothing vapor permeation efficiency for a given activity and airflow (N.D.)                   | $\varepsilon_{cl}$ | Emissivity of the outer surface of the clothing (N.D.)                                                              |
| $I_{cl}$            | Intrinsic clothing insulation for a given activity and airflow (clo)                           | $\eta$             | Sweat evaporation efficiency (N.D.)                                                                                 |
| $i_{cl, st}$        | Static clothing vapor permeation efficiency (N.D.)                                             | $\sigma$           | Stefan-Boltzmann constant (W/(m <sup>2</sup> K <sup>4</sup> ))                                                      |
| $I_{cl, st}$        | Static intrinsic clothing insulation (clo)                                                     |                    |                                                                                                                     |

of the body deteriorates, and the possibility of heatstroke increases. When heatstroke occurs, the body cannot regulate its temperature, and the core body temperature increases. This situation results in mental confusion and loss of consciousness, requiring urgent medical treatment for recovery [4]. Because thermal regulation controls the core body temperature and heatstroke results thermal regulation failure, the core body temperature is a critical parameter for evaluating the risk of heatstroke. The International Organization for Standardization (ISO) defines the upper limit of the acceptable rectal temperature limit as 38.0 °C [5]; the American Conference of Governmental Industrial Hygienists (ACGIH) defines it as 38.0 °C for unacclimatized workers and 38.5 °C for selected and acclimatized workers [6]. When an esophageal temperature is monitored and medical surveillance and emergency treatment are simultaneously provided, ISO permits an upper limit of 39.0 °C for acclimatized workers [5]. With this limit, to maintain body thermal equilibrium, the Heat Strain Decision Aid (HSDA) model [7] proposed for army use adopts 38.3 °C when the evaporation capacity in the evaluated environment is lower than the required evaporative cooling and 39.5 °C when the evaporation capacity is higher than this value. These temperatures have been updated to 38.5 and 40.0 °C, respectively, in the Army Technical Bulletin Medical 507 [8]. Heatstroke is unavoidable when workers work continuously in an environment hotter than the prescriptive zone, even when they ingest sufficient water and salt. Therefore, evaluating whether the surrounding environment meets the requirements of the prescriptive zone is worthwhile.

The predicted mean vote (PMV) [2] enables the prediction of whether the combination of the type of activity and the surrounding environment, including clothing, ensures a thermally comfortable state or if the combination leads to a warmer or cooler sensation than that corresponding to thermal neutrality. Because skin temperature and sweat rate are critical parameters for thermal comfort assessment, the PMV is derived from the analysis of the heat balance of the human body based on the boundary conditions for comfortable skin temperature and sweat rate. If the heat balance of the human body is established, the PMV considers that the combination leads to a thermally neutral state corresponding to thermal comfort. If the heat dissipation surpasses the body heat generation, the PMV predicts that the combination causes a cooler feeling than the neutral state. If the heat dissipation is lower than the body heat generation, the PMV indicates that the combination causes a warmer sensation. The PMV has been adopted as the standard index to evaluate thermal comfort [9,10]. However, although the core body temperature is a critical parameter for assessing the risk of heatstroke, a simple equation that describes the heat interaction between the core body, skin, and surrounding environment, including clothing, to evaluate whether heat storage occurs in the core body is currently lacking. To address this shortcoming, an equation corresponding to PMV is proposed for heatstroke risk evaluation in this study.

The wet-bulb globe temperature (WBGT) was developed to evaluate the heat stress and heat disorder risks [11]. The WBGT is a simplified index of the effective temperature proposed by Houghton and Yaglou

[12] to express the threshold for heat disorder incidence. The WBGT is the weighted average of the following environmental temperatures: natural wet-bulb temperature ( $T_{nw}$ ), globe temperature ( $T_g$ ), and air temperature ( $T_a$ ).  $T_{nw}$ ,  $T_g$ , and  $T_a$  theoretically coincide with the WBGT in a uniform temperature environment at 100 % relative humidity (RH). Therefore, the WBGT can be interpreted as the perceived temperature expressed in a uniform temperature environment at 100 % RH. It effectively evaluates the influence of environmental temperature and humidity on workers wearing general vapor-permeable clothing [13]. The WBGT is widely used because of its validity and ease of in-field determination [4,6,14,15]. However, when clothing properties diverge from those of general work clothes, the WBGT needs to be adjusted suitably to refer to the original WBGT risk scale [4,6,14,15]. This adjustment is required because the WBGT, as an environmental heat stress index, is not always consistent with the body heat strain, such as the core body temperature, and body dehydration. The WBGT is considered a risk-screening index for on-site heat disorders. When a more detailed risk assessment is required, the use of the predicted heat strain (PHS) [16] is recommended [17].

To evaluate the heat stress or heat strain, human heat balance models, such as the PHS, heat stress index (HSI) [18], and required sweat rate (SWreq) [19], have been proposed. The HSI calculates the required sweat evaporation to maintain the body thermal equilibrium based on the heat balance of the human body with skin temperature at 35 °C. The HSI does not include the core body temperature in its calculation. The heat stress is expressed as the ratio of the required evaporation to the evaporation capacity in the evaluated environment. The SWreq also calculates the sweat rate required to maintain thermal equilibrium based on the heat balance of the human body, but it considers non-evaporative sweat (dripping sweat) and skin temperature variations depending on the metabolic rate and the surrounding environment as a multiple regression formula in its calculation process. The core body temperature is not included in the SWreq calculation. The SWreq evaluates the duration of limited exposure (DLE) based on body dehydration and heat storage. However, the skin temperature and sweat rate of the SWreq do not change over time. The ISO previously adopted the SWreq as the standard for evaluating heat stress [19], which was then replaced with the PHS [17]. To calculate the PHS, the heat transfer between the body core, skin, and surrounding environment is analyzed, and the transient core body temperature, skin temperature, and sweat rate are calculated. The DLE is then determined as the time required to reach body dehydration or the core body temperature limit. Time constants are adopted to express the time delays for the target core body, skin temperatures, and sweating. Both the target core body and skin temperatures are the set-point temperatures for determining the sweat rate required to maintain thermal equilibrium. Non-evaporative sweat (dripping sweat) is considered and a 1 min calculation time step is applied in the calculation. The skin temperature in the PHS calculation is changed to the target skin temperature without a time delay, whereas the sweat rate is changed to the required sweat rate with a time delay, expressed as the time constant for sweating. The core body temperature is determined to satisfy the heat balance for a body with the above skin temperature and sweat rate. The advantage of the PHS model is that it can predict the transient core body temperature, skin temperature, and sweat rate. Wang et al. reported that their PHS predictions agreed with the thermal physiological responses of subjects wearing general clothing but diverged from those of subjects wearing protective clothing [20]. As the human thermal model has become increasingly complex, it has gained the ability to predict additional parameters. However, the model accuracy deteriorates if the setting parameters are inadequate for the evaluated condition, such as in the case of the protective clothing reported by Wang et al. [20]. A simple equation with the least parameters describing heatstroke risk, comparable to PMV used for thermal comfort evaluation, is required.

Several human thermal models have been developed to predict the transient response of the human body, such as those of Stolwijk [21],

Gagge et al. [22], Jones and Ogawa [23], Fiala et al. [24], Huizenga et al. [25], Kohri and Mochida [26], Ooka et al. [27], Kurazumi et al. [28], and Takahashi et al. [29]. However, these models require many more parameters than the PMV. Although the applicability of the PMV is limited to thermally neutral and cool states [30], it provides practical information for judging whether the thermal environment is comfortable for occupants residing in buildings with heating, ventilation, and air conditioning systems [9,10]. The simplicity of the PMV enables users to interpret the output clearly. Many scholars have evaluated heat strain, and some of them have used human thermal models such as the PHS [17, 31,32], HSDA [7,33,34], and Fiala model [24,35]. Human trials have been also conducted by various authors, including Bernerd et al. [13], Wang et al. [20], Kopeckova et al. [35], and Mori et al. [36]; moreover, sweating thermal manikin measurements [34,37] have been performed. These researchers evaluated the transient human body responses in given conditions. However, none of them provided a simple heat balance equation for assessing the risk of heatstroke that is comparable to the PMV for the assessment of thermal comfort.

In the present paper, we refer to the threshold as the boundary limit for the combination of the parameters of activity, clothing, and the surrounding environment for heatstroke occurrence. We illustrate the human heat interactions between the body core, skin, clothing, and indoor and outdoor environments, and based on this illustration, we propose a simple heat balance equation, comparable to that for the PMV used for thermal comfort evaluation. This equation enables users to predict whether the combination of activity, clothing, and the surrounding environment can cause the threshold to be exceeded, i.e., can cause the body temperature to rise above the acceptable temperature.

## 2. Methods

We propose the target heat balance equation based on the modelling of heat interaction between the body core, skin, clothing, and surrounding environment and setting parameters and formulas for the model referring to the existing studies and standards.

First, we illustrate the heat interactions between the body core, skin, clothing, and surrounding environment. We represent the human body as two parts: core and shell. The outer surface of the shell is the skin. The driving forces of heat flow are the temperature difference and equilibrium temperature, and increases and decreases in the temperature of the considered body are the results of heat inflow, heat outflow, and heat generation. These processes are similar to water flows and water heads because water flow is caused by a water head difference, and water head is governed by water inflow, outflow, and gushing out. Therefore, we mimic heat interactions using water flows and water heads in the illustration. Because water evaporation is heat transfer accompanied by a volume change from liquid to gas, evaporation is represented by a heat pump, which also utilizes volume change for heat transfer in this illustration.

As an autonomous thermal regulator, the human body changes the heat transfer characteristics inside the body. The human body contains subcutaneous fat located between the body core and skin. Because the thermal conductivity of subcutaneous fat is low, it provides thermal insulation to the body. The thermal insulation effect of subcutaneous fat is altered by the penetration of skin blood flow into it. Vasoconstriction of skin blood vessels and reduced blood flow maintain the thermal insulation effect of the subcutaneous fat and keep the core body temperature high when the skin temperature is low. Vasodilation of the skin blood vessels and increased blood flow, including that in arteriovenous anastomoses, reduces the effect of the thermal insulation provided by the subcutaneous fat and enhances heat loss from the body core to the skin when the skin temperature is high. Sweating encourages body heat loss by evaporation, whereas shivering produces heat to prevent a temperature drop. Because the evaporation of sweat provides a strong cooling effect (approximately 2400 J/ml), the sweat rate must be well controlled to avoid an excessive drop in body temperature. Sweat

secretion is not observed in motionless occupants in thermally neutral states. In contrast, it occurs in active occupants or in those exposed to warmer conditions.

In behavioral thermal regulation, humans change clothing and the surrounding environment by operating a fan or an air conditioner. Air conditioning adjusts the temperature difference between the indoor environment and skin. Changing clothes modifies the heat and vapor transfer characteristics between the skin and outer surface of the clothing. A fan in operation changes the heat and vapor transfer characteristics between the outer surface of the clothing and the surrounding environment. Here, the operative temperature of the indoor environment is referred to as the temperature of the surrounding environment. The outdoor environment is an influential factor that affects the indoor operative temperature. The degree of outdoor influence depends on factors such as building insulation, opening and closing of windows and doors, and ventilation.

Through these processes, both the core body and skin temperatures are affected by clothing, airflow around the human body, and indoor and outdoor environments. These interactions are illustrated in Fig. 1, which shows the conditions wherein the outdoor temperature is lower than the indoor temperature, and the indoor temperature is lower than the skin temperature.

The temperature is indicated as the water head height in Fig. 1. The upper and lower sides correspond to high and low temperatures, respectively. As mentioned above, the heat flows are expressed as water flows caused by the water head difference, that is, the temperature difference. Because the inflow–outflow balance and tank cross-sectional area determine the changing speed of the water head, the cross-sectional area of the water tank can be interpreted as the heat capacity of the component corresponding to the water head. Valve control and heat pump operation represent the autonomous and behavioral thermal regulations to maintain the core body temperature ( $T_{cr}$ ) within a narrow range by balancing the metabolic heat and heat dissipation from the core body. The difference between  $T_{cr}$  and the skin temperature ( $T_{sk}$ ) causes heat flow from the core body to the skin, which is controlled by the skin blood flow. The difference between  $T_{sk}$ , the clothing outer surface temperature ( $T_{cl}$ ), and the indoor operative temperature ( $T_{op}$ ) causes heat to flow from the skin to the indoor environment through the clothing. When the heat flow from the skin to the indoor environment is insufficient,  $T_{sk}$  and  $T_{cr}$  increase, and sweat secretion, which acts as a heat pump, occurs to maintain  $T_{cr}$  constant.

Adjustments to clothing and airflow modify the heat transfer characteristics between the skin and the indoor environment. Opening and

closing of windows and doors, ventilation, and operating an air conditioner (heating and cooling) change  $T_{op}$ . Fig. 1 was drawn by assuming that  $T_{sk}$  is higher than  $T_{op}$  and  $T_{op}$  is higher than the outdoor operative temperature ( $T_{out}$ ). However, these relationships may be the opposite. When  $T_{op}$  is higher than  $T_{sk}$ , sweat secretion maintains  $T_{sk}$  at a lower level than  $T_{op}$  through sweat evaporation. When  $T_{out}$  is higher than  $T_{op}$ , an air conditioner is operated to pump heat from the indoor to the outdoor environment to maintain  $T_{op}$  lower than  $T_{out}$ . As described in Section 1,  $T_{sk}$  and sweat rate are critical factors for thermal comfort assessment. Simultaneously,  $T_{cr}$  is critical for assessing the risk of heatstroke. Because several valves and heat pumps shown in Fig. 1 affect the water heads of  $T_{sk}$  and  $T_{cr}$ , an integrated design dealing with clothing, airflow around the human body, and indoor and outdoor environments as a unified system is effective in achieving a comfortable  $T_{sk}$  or acceptable  $T_{cr}$ .

Based on the illustrated heat interaction between the body and surrounding environment in Fig. 1, we propose a heat balance equation that estimates whether a given combination of activity (metabolic rate), clothing, air temperature, mean radiant temperature, air velocity, and vapor pressure exceeds the threshold for heatstroke occurrence, that is, whether the combination increases  $T_{cr}$  beyond its upper limit ( $T_{cr,lim}$ ). Because  $T_{cr}$  in this study is defined for the case where the body is divided into two parts, the core body and shell,  $T_{cr}$  is not as precise as that monitored using a sensor set in the esophageal tract. We set  $T_{cr,lim}$  to 38.5 and 38.0 °C for acclimatized and unacclimatized workers, respectively, by referring to an ACGIH set for rectal temperature [6]. Because we focused on the upper limit of  $T_{cr}$ , the sweat rate was assumed to reach the maximum ( $SW_{max}$ ). The following formulas suggested by the ISO [17] were adopted.

For unacclimatized workers,

$$SW_{max} = 1.8(M - 32), \text{ provided } 250 \leq SW_{max} \leq 400, \quad (1)$$

and for acclimatized workers,

$$SW_{max} = 2.25(M - 32), \text{ provided } 312.5 \leq SW_{max} \leq 500, \quad (2)$$

where  $SW_{max}$  is the maximum sweat rate expressed as the latent heat per unit body surface area ( $W/m^2$ ), and  $M$  is the metabolic rate ( $W/m^2$ ).

The thermal resistance between the core body and skin ( $R_{cs}$ ) for the case in which the skin blood flow reaches its maximum ( $R_{cs,lim}$ ) needs to be determined. We calculated the thermal resistance between the core body and skin ( $R_{cs}$ ) using Eq. (3) by referring to the steady-state data reported by Wang et al. [38] for unacclimatized workers and by Frye and

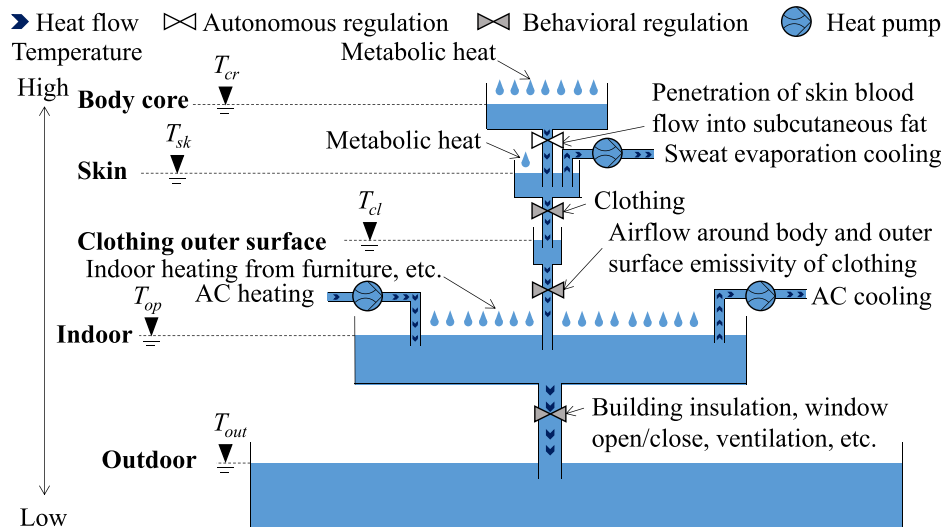


Fig. 1. Interactions between the human body and surrounding environment through heat transfer.



Kamon [39] for acclimatized workers:

$$R_{cs} = \frac{T_{cr} - T_{sk}}{M - C_{res} - E_{res}}, \quad (3)$$

where  $R_{cs}$  is the thermal resistance between the core body and skin ( $\text{m}^2\text{C/W}$ );  $T_{cr}$  is the core body temperature ( $^{\circ}\text{C}$ );  $T_{sk}$  is the skin temperature ( $^{\circ}\text{C}$ ); and  $C_{res}$  and  $E_{res}$  are the sensible and latent heat losses through respiration, respectively ( $\text{W/m}^2$ ).

The calculated  $R_{cs}$  ranged from 0.013 to 0.040  $\text{m}^2\text{C/W}$  for unacclimatized workers and from 0.010 to 0.018  $\text{m}^2\text{C/W}$  for acclimatized workers. We adopted  $R_{cs,\min}$  of 0.0145  $\text{m}^2\text{C/W}$  and 0.012  $\text{m}^2\text{C/W}$  for unacclimatized and acclimatized workers, respectively, as the values at which the skin blood reaches the maximum to reduce  $R_{cs}$ . We adopted a slightly larger value of  $R_{cs,\min}$  than the smallest calculated  $R_{cs}$  for risk reduction. This equation distinguishes between unacclimatized and acclimatized workers by changing  $T_{cr,\lim}$ ,  $SW_{\max}$ , and  $R_{cs,\min}$ .

We adopted the formula for evaporation sweat efficiency ( $\eta$ ) given for SWreq [19] and the formulas for sensible and latent heat losses through respiration ( $C_{res}$  and  $E_{res}$ , respectively) provided by the ISO [17]. Then, if the air temperature ( $T_a$ ), mean radiant temperature ( $T_r$ ), air velocity ( $v$ ), intrinsic clothing insulation ( $I_{cl}$ ), clothing vapor permeation efficiency ( $i_{cl}$ ), vapor pressure  $P_a$ , metabolic rate ( $M$ ), and external work ( $W$ ) are given, we can calculate the total steady heat loss  $Q$  from the core body having  $T_{cr,\lim}$ ,  $R_{cs,\min}$ , and  $SW_{\max}$  by combining Eqs. (4)–(10). Eq. (8) is a transformation of Antoine's equation [40] into a linear equation:

$$Q = \frac{T_{cr,\lim} - T_{sk}}{R_{cs,\min}} + C_{res} + E_{res} = \frac{T_{sk} - T_{cl}}{0.155I_{cl}} + E_{sk} + C_{res} + E_{res}, \quad (4)$$

$$\frac{T_{sk} - T_{cl}}{0.155I_{cl}} = h_c(T_{cl} - T_a)f_{cl} + h_r(T_{cl} - T_r)f_{cl}, \quad (5)$$

$$h_r = \epsilon_{cl}\sigma[(T_{cl} + 273.15)^2 + (T_r + 273.15)^2][(T_{cl} + 273.15) + (T_r + 273.15)]f_{cl}, \quad (6)$$

$$E_{\max} = LRh_cF_{pct}(P_{sk}^* - P_a)f_{cl}, \quad (7)$$

provided that

$$F_{pct} = \frac{1}{1 + \frac{0.155I_{cl}h_c f_{cl}}{i_{cl}}}.$$

$$P_{sk}^* = \alpha T_{sk} + \beta, \quad (8)$$

provided that

$$\alpha = \frac{4030.183}{(235 + T_{sk})^2} e^{\left(16.6536 - \frac{4030.183}{235 + T_{sk}}\right)}$$

$$\beta = \left(1 - \frac{4030.183T_{sk}}{(235 + T_{sk})^2}\right) e^{\left(16.6536 - \frac{4030.183}{235 + T_{sk}}\right)}$$

$$E_{sk} = \eta SW_{\max} = wE_{\max}, \quad (9)$$

$$\eta = \begin{cases} 1 - \frac{w^2}{2} \dots \frac{E_{\max}}{SW_{\max}} > \frac{1}{2} \\ \frac{E_{\max}}{SW_{\max}} \dots \frac{E_{\max}}{SW_{\max}} \leq \frac{1}{2} \end{cases}. \quad (10)$$

In the above equations,  $Q$  is the total steady heat loss from the core body having  $T_{cr,\lim}$ ,  $R_{cs,\min}$ , and  $SW_{\max}$  ( $\text{W/m}^2$ );  $R_{cs,\min}$  is  $R_{cs}$  for a human body with the maximum skin blood flow ( $\text{m}^2\text{C/W}$ );  $T_{cl}$  is the outer surface temperature of the clothing ( $^{\circ}\text{C}$ );  $I_{cl}$  is the intrinsic clothing insulation for a given activity and airflow ( $\text{clo}$ );  $E_{sk}$  is the evaporative heat loss ( $\text{W/m}^2$ );  $h_c$  and  $h_r$  are the convective and radiative heat transfer coefficients, respectively ( $\text{W/(m}^2\text{C)}$ );  $T_a$  is the air temperature ( $^{\circ}\text{C}$ );  $f_{cl}$  is

the clothing area factor (N.D.);  $T_r$  is the mean radiant temperature ( $^{\circ}\text{C}$ );  $\epsilon_{cl}$  is the emissivity of the outer surface of the clothing (N.D.);  $\sigma$  is the Stefan–Boltzmann constant ( $\text{W/(m}^2\text{K}^4)$ );  $F_r$  is the radiation area factor of the human body (N.D.);  $E_{\max}$  is the evaporation capacity of the surrounding environment including the clothing ( $\text{W/m}^2$ );  $LR$  is the Lewis ratio ( $\text{K/kPa}$ );  $F_{pct}$  is the vapor permeation efficiency (N.D.);  $P_{sk}^*$  is the saturated vapor pressure at  $T_{sk}$  ( $\text{kPa}$ );  $P_a$  is the environmental vapor pressure ( $\text{kPa}$ );  $i_{cl}$  is the clothing vapor permeation efficiency for a given activity and airflow (N.D.);  $\alpha$  is the differential coefficient of Antoine equation by  $T_{sk}$  ( $\text{kPa}/^{\circ}\text{C}$ );  $\beta$  is the intercept of the linear equation around  $T_{sk}$  for the Antoine equation ( $\text{kPa}$ ); and  $\eta$  is the sweat evaporation efficiency;  $w$  is the skin wettedness (N.D.).

We evaluate the heat stored in the human body core under maximum thermal regulation ( $S$ ) by applying Eq. (11):

$$S = M - W - Q = M - W - \left( \frac{T_{cr,\lim} - T_{sk}}{R_{cs,\min}} + C_{res} + E_{res} \right) = M - W - \left( \frac{T_{sk} - T_{cl}}{0.155I_{cl}} + E_{sk} + C_{res} + E_{res} \right) = M - W - [h_c(T_{cl} - T_a)f_{cl} + h_r(T_{cl} - T_r)f_{cl} + wE_{\max} + C_{res} + E_{res}], \quad (11)$$

where  $S$  is a heat stored in the human body core under maximum thermal regulation ( $\text{W/m}^2$ ).

If  $S$  is greater than zero, the given combination of activity, clothing, air temperature, mean radiant temperature, air velocity, and vapor pressure exceeds the threshold for the occurrence of heatstroke. Thus, there is a risk of heatstroke for workers engaged in continuous work because heat is accumulated in the body core and  $T_{cr}$  exceeds  $T_{cr,\lim}$ . If  $S$  is equal to or less than zero, the given combination is lower than the threshold. The risk of heatstroke is low if workers ingest sufficient amounts of water and salt. We propose Eq. (11) as a heatstroke threshold equation, which is solved referring to Eqs. (1), (2) and (4)–(10). The analysis flow of Eq. (11) will be given later.

As the integrated relationships in Eqs. (4)–(11) are difficult to imagine, we illustrate these relationships in Fig. 2.

In Fig. 2,  $P_{sk}$  is the vapor pressure at skin ( $\text{kPa}$ ).  $P_{sk}$  is the weighted average of  $P_{sk}^*$  and  $P_a$  by  $w$  and  $(1 - w)$ .

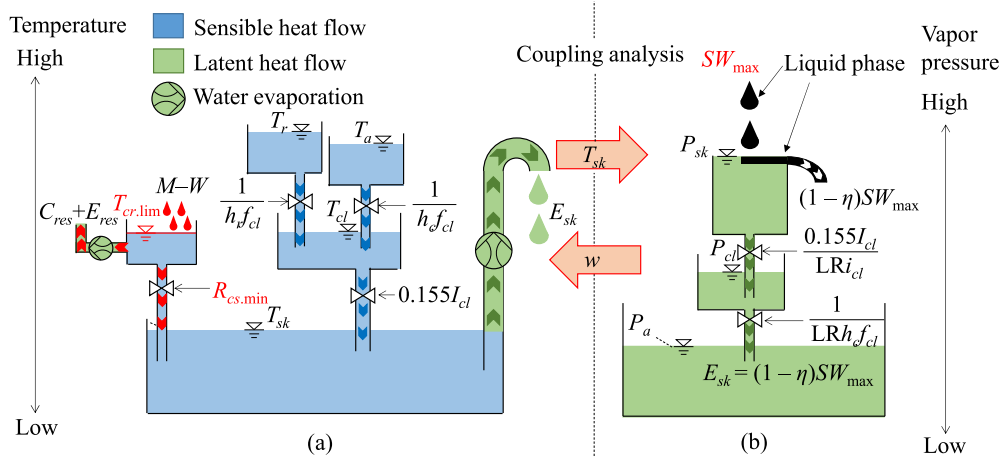
Fig. 2 assumes that  $T_a$  and  $T_r$  are higher than  $T_{cr,\lim}$ , and that the metabolic heat is dissipated to the surrounding environment through sweat evaporation, which acts as a heat pump for the human body. For the thermal physiological parameters, we set  $T_{cr,\lim}$ ,  $R_{cs,\min}$ , and  $SW_{\max}$  (depicted in red in Fig. 2) as the parameters for the maximum thermal regulation against heat.  $T_{sk}$  can be determined uniquely because we can identify  $C_{res}$ ,  $E_{res}$ , and all the valve resistances corresponding to the heat and moisture transfers shown in Fig. 2.  $Q$  is obtained by substituting the identified  $T_{sk}$  into Eq. (4). The comparison between the inflow to the body core ( $M - W$ ) and the outflow from it ( $Q$ ) enables judgment of whether  $S$  occurs in the body core having  $T_{cr,\lim}$  (this situation corresponds to the rise in  $T_{cr}$  from  $T_{cr,\lim}$ , indicated as the red water head in Fig. 2).

In the analysis, we solved  $T_{sk}$  using a provisional set of  $\alpha$  and  $\beta$  for a provisional  $T_{sk}$ . Then,  $\alpha$  and  $\beta$  were updated by referring to the solved  $T_{sk}$ . The calculation was iterated until  $\alpha$  and  $\beta$  converged. The detailed steps are as follows.

Step 1: Set  $T_{sk}$  and  $T_{cl}$  tentatively, and calculate  $\alpha$ ,  $\beta$ , and  $h_r$  using Eqs. (6) and (8).

Step 2: Using the tentative  $T_{sk}$  and the calculated  $h_r$ , update  $T_{cl}$  based on Eqs. (5) and (6). Next, obtain the converged  $T_{cl}$  and  $h_r$  by iterating steps 1 and 2.

Step 3: Assuming  $w$  is unity, i.e.,  $E_{\max} \leq 0.5SW_{\max}$ , solve  $T_{sk}$  using Eqs. (4), (5) and (7)–(9). By iterating steps 1–3, search for the converged  $T_{sk}$ ,  $T_{cl}$ , and  $h_r$ . Then, compare  $E_{\max}$  and  $0.5SW_{\max}$ . If  $E_{\max} \leq 0.5SW_{\max}$ , use the obtained values and move to step 8. If not, proceed to step 4 for the case in which  $w$  is less than unity, i.e.,  $E_{\max} > 0.5SW_{\max}$ .



**Fig. 2.** Illustration of relationships between Eqs. (4)–(10) and outline of the heatstroke equation: (a) whole body heat balance and (b) liquid sweat and sweat evaporation heat flow.

Step 4: Repeat step 1.

Step 5: Repeat step 2.

Step 6: Solve  $w$  by combining Eqs. (7)–(10). Here, we refer to Eq. (10) for the condition  $E_{\max} > 0.5SW_{\max}$ . The obtained  $w$  is substituted into Eq. (9) to calculate  $E_{sk}$ .

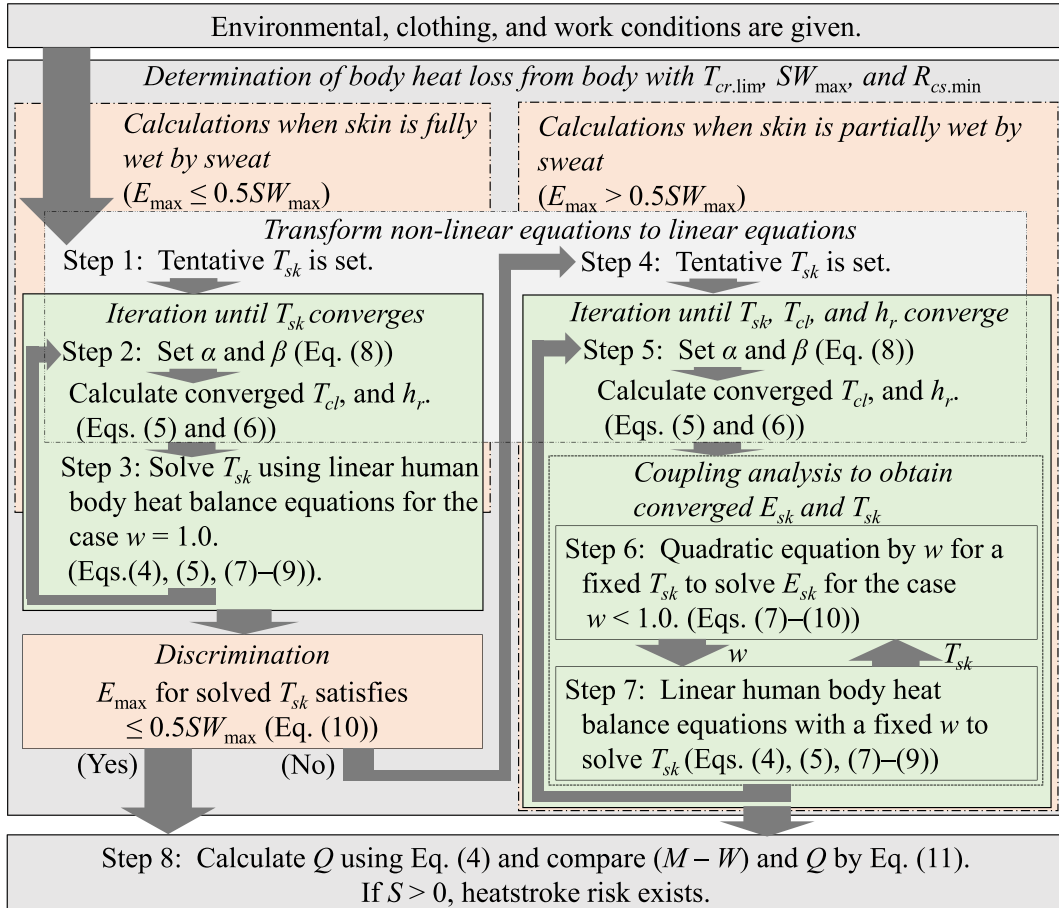
Step 7: Assuming a constant  $w$ , solve  $T_{sk}$  by combining Eqs. (4), (5), (7)–(9). By iterating steps 4–7, search for the converged  $T_{sk}$ ,  $w$ ,  $T_{cl}$ , and  $h_r$ .

Step 8: Calculate  $Q$  as the sum of  $C_{res}$ ,  $E_{res}$ , and  $(T_{cr,lim} - T_{sk})/R_{cs,min}$ . Then, compare  $(M - W)$  and  $Q$  referring to  $S$  in Eq. (11).

These calculation flows are summarized in Fig. 3.

### 3. Results

We analyzed an isothermal environment in which an occupant is



**Fig. 3.** Summary of analysis flow.

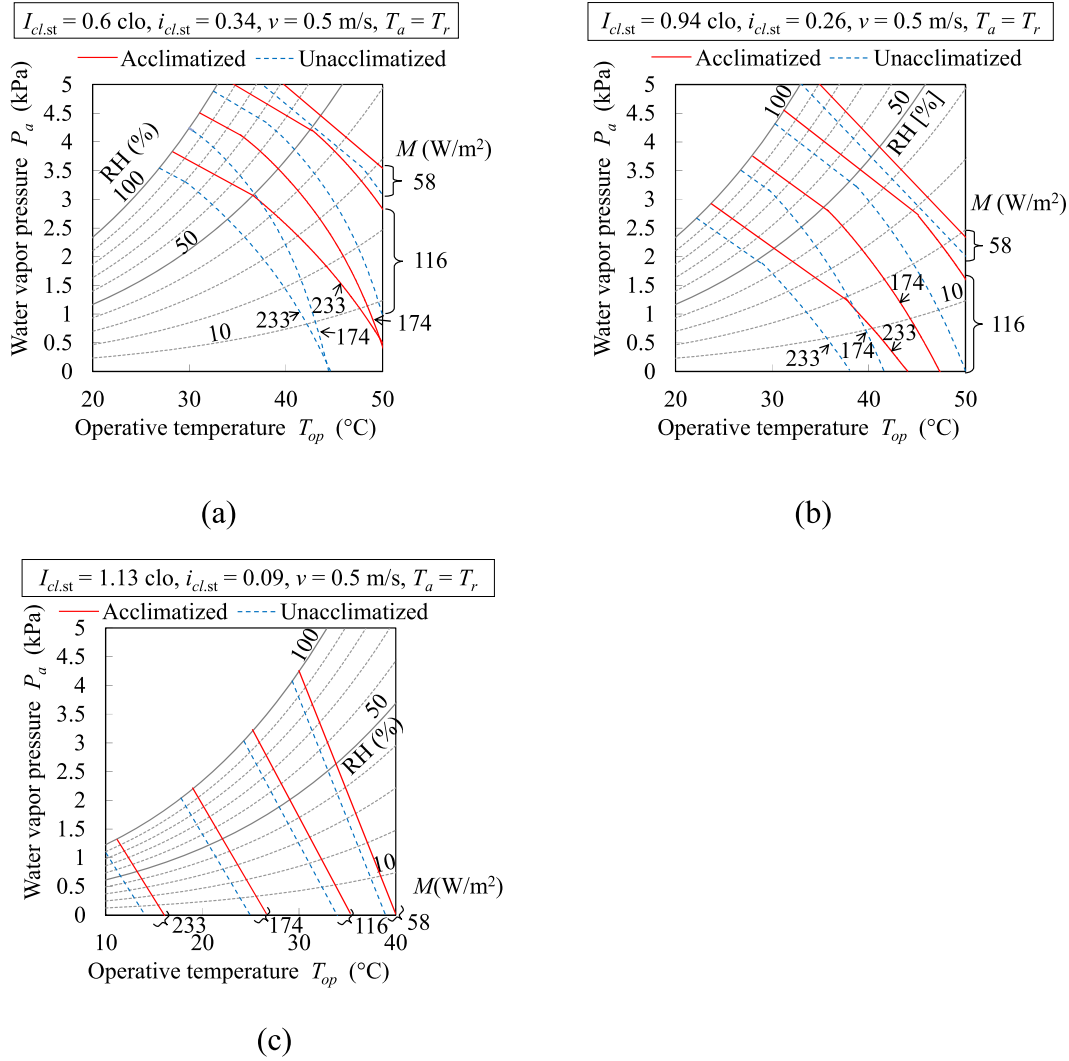


Fig. 4.  $T_{op}$  and  $P_a$  of thresholds for three clothing ensembles for various activities: (a) WC, (b) VPAIP, and (c) FLP.

walking on a treadmill. Because the occupant remains in the same place, we set the air velocity of the surrounding environment to the air velocity relative to the human body. The convective heat transfer coefficient ( $h_c$ ) was calculated using the formula for the standing posture proposed by de Dear et al. [41]. Before the calculations, we converted the static intrinsic clothing insulation ( $I_{cl, st}$ ) and static clothing vapor permeation efficiency ( $i_{cl, st}$ ) into  $I_{cl}$  and  $i_{cl}$ , respectively, according to the ISO [42] and by referring to Sakoi and Mochida [43]. These variables vary with body motion and ambient airflow. Because the walking speeds at 3 and 4 met ( $M$  of 174 and 233  $W/m^2$ ) exceeded the input range for the formulas suggested by the ISO [42], the corresponding calculations were considered preliminary results. In this section, we discuss the influence of several factors on the threshold of heatstroke occurrence based on the effect of each of the factors on  $T_{op}$  of the threshold. The calculation conditions reported in this section are summarized in Table 1.

Fig. 4 describes the combinations of  $T_{op}$  and  $P_a$  of the thresholds for three clothing ensembles: work clothes (WC), vapor-permeable but air-impermeable protective clothing (VPAIP), and film-laminated protective clothing (FLP). We set  $I_{cl, st}$  of 0.6 clo and  $i_{cl, st}$  of 0.34 N.D. for WC [42]. We set  $I_{cl, st}$  to 0.95 clo and  $i_{cl, st}$  to 0.26 N.D. for VPAIP by referring to the values for Tyvek coverall [42]. The corresponding values for FLP were set as 1.13 clo and 0.09 N.D. by referring to the values for PVC and vinyl acid suit [42]. The results in Fig. 4 correspond to the conditions of air velocity ( $v$ ) = 0.5 m/s and  $M$  between 58 and 233  $W/m^2$ .

The thresholds for each metabolic rate were drawn to connect the

high  $P_a$  and low  $T_{op}$  regions with the low  $P_a$  and  $T_{op}$  high regions. The straight lines with a negative slope correspond to the state with  $w$  at unity, and the upward convex curves with a negative slope correspond to the state with  $w$  less than unity.  $w$  decreases with decreasing  $P_a$  for each convex curve. All thresholds for acclimatized workers are located in the higher  $T_{op}$  and  $P_a$  regions than those for unacclimatized workers for the same activity level. The gentle slopes of the thresholds indicate that  $P_a$  is the primary determinant of the thresholds. Thus, we can say that the influence of  $P_a$  on the threshold decreases with a shift to lower  $P_a$ . The slopes of the thresholds for FLP (Fig. 4(c)) are steeper than those for the WC (Fig. 4(a)) and VPAIP (Fig. 4(b)). This difference is attributed to the higher evaporative heat transfer resistance of clothing for FLP than for WC and VPAIP because of the higher  $I_{cl}$  and lower  $i_{cl}$  of FLP. Except for the low  $P_a$  environment for WC (Fig. 4(a)), the increase in work intensity, that is, the metabolic rate, shifts the threshold to the low  $T_{op}$  environment. The higher  $SW_{max}$  of Eqs. (1) and (2) at an  $M$  value of 233  $W/m^2$  than that at 174  $W/m^2$  may explain the opposite tendency of the threshold curve in the exception, which could change the evaporative cooling to a greater extent than the difference in heat production.

Fig. 5 shows the combination of  $T_{op}$ ,  $v$ , and  $M$  at the threshold at  $P_a$  of 3.4 kPa (60 % RH at 35 °C).

The thresholds were drawn as curves connecting the low  $T_{op}$  and  $v$  regions, where the slope of the curve is moderate, to the high  $T_{op}$  and  $v$  regions, where the slope becomes steep. All thresholds for acclimatized workers are located in higher  $T_{op}$  regions than those for unacclimatized

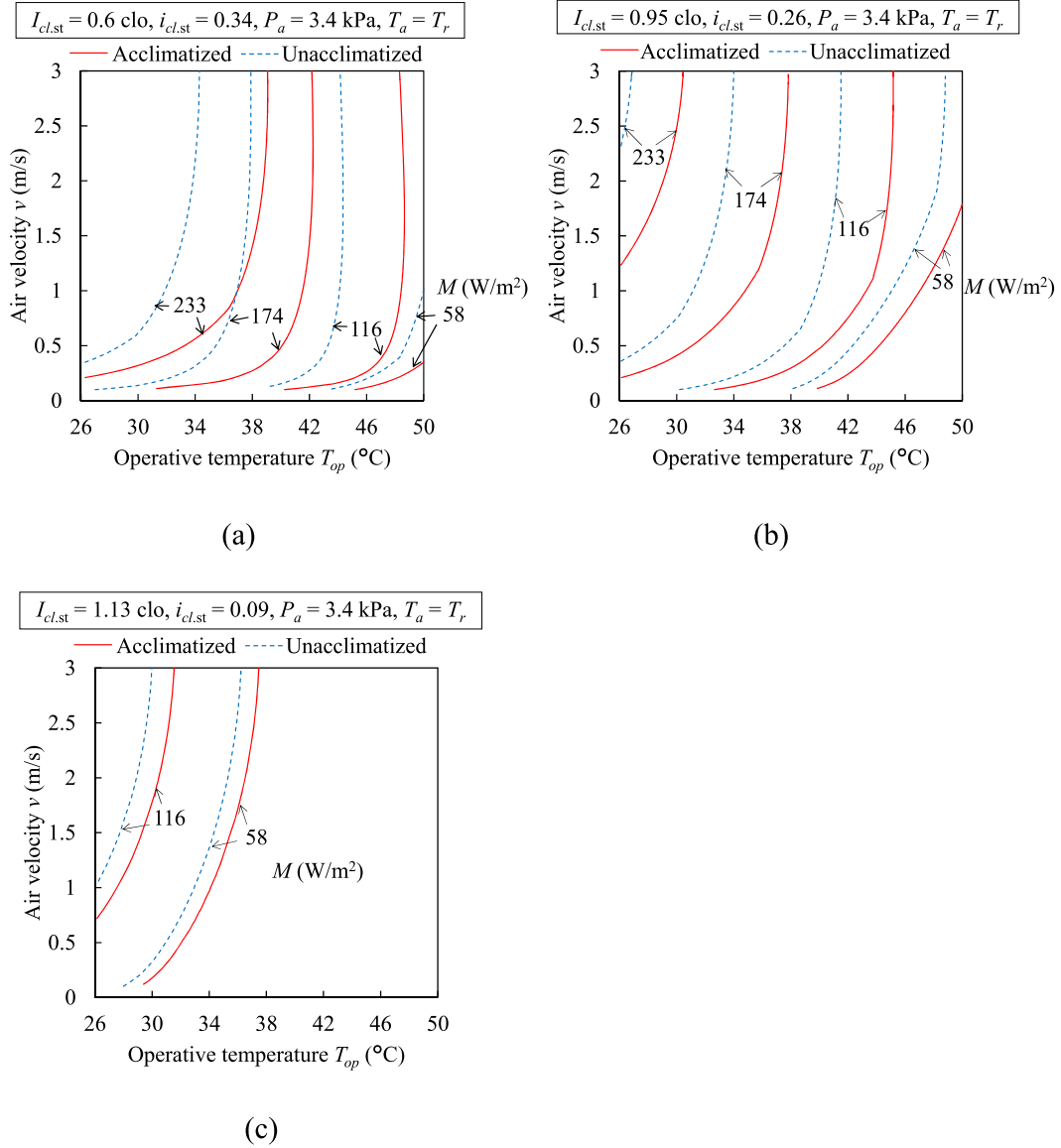


Fig. 5. Combinations of  $T_{op}$ ,  $v$ , and  $M$  at the threshold at  $P_a$  of 3.4 kPa for three clothing ensembles: (a) WC, (b) VPAIP, and (c) FLP.

workers at the same activity level. Overall, the increase in  $i_{cl,st}$  and decrease in  $I_{cl,st}$  move all thresholds at the same activity level to the higher  $T_{op}$  region. For thresholds within the range indicated in Fig. 5,  $w$  is maintained at unity under the conditions of  $M$  at 58 W/m<sup>2</sup> for acclimatized workers with WC, 233 W/m<sup>2</sup> for both acclimatized and unacclimatized workers with VPAIP, and for all conditions related to FLP. Except for these conditions,  $w$  is unity only in a low  $v$  environment and decreases with an increase in  $v$ . At specific points, the slope of the curve changes sharply. At these points,  $w$  changes from 1 to less than 1. When  $v$  is lower than the value at such points,  $T_{op}$  of the threshold changes sensitively with  $v$ . This characteristic implies that small values of  $v$  influence the threshold. According to Eq. (10),  $\eta$  is less than 0.5 when  $w$  is maintained at unity. Under this condition, the increase in  $v$  increases  $E_{max}$  and  $\eta$  intensively and enhances the evaporative cooling. Therefore, small values of  $v$  substantially affect the threshold. When  $v$  is high,  $E_{max}$  is also high, and  $\eta$  is close to unity. Thus, the increase in  $v$  brings a slight improvement in  $\eta$ , and the evaporative cooling is not enhanced significantly. In contrast, the heat inflow from the surrounding environment to the human body increases with an increase in  $v$ . Because the increase in heat inflow compensates for the increase in evaporative cooling, the effect of  $v$  on the thresholds is not apparent. Regarding the influence of

$i_{cl,st}$ , in the case of clothing with low  $i_{cl,st}$ , such as FLP,  $w$  remains at unity, and  $E_{max}$  becomes the determinant factor for  $\eta$  given by Eq. (10). As  $E_{max}$  increases gradually with both low and high values of  $v$ , the thresholds in environments with both high and low values of  $v$  are moderately affected by it.

Fig. 6 indicates the combinations of  $I_{cl,st}$  and  $i_{cl,st}$  that cause the same threshold  $T_{op}$  for the condition with  $v$  at 0.5 m/s and  $P_a$  at 3.4 kPa (60 % RH at 35 °C).

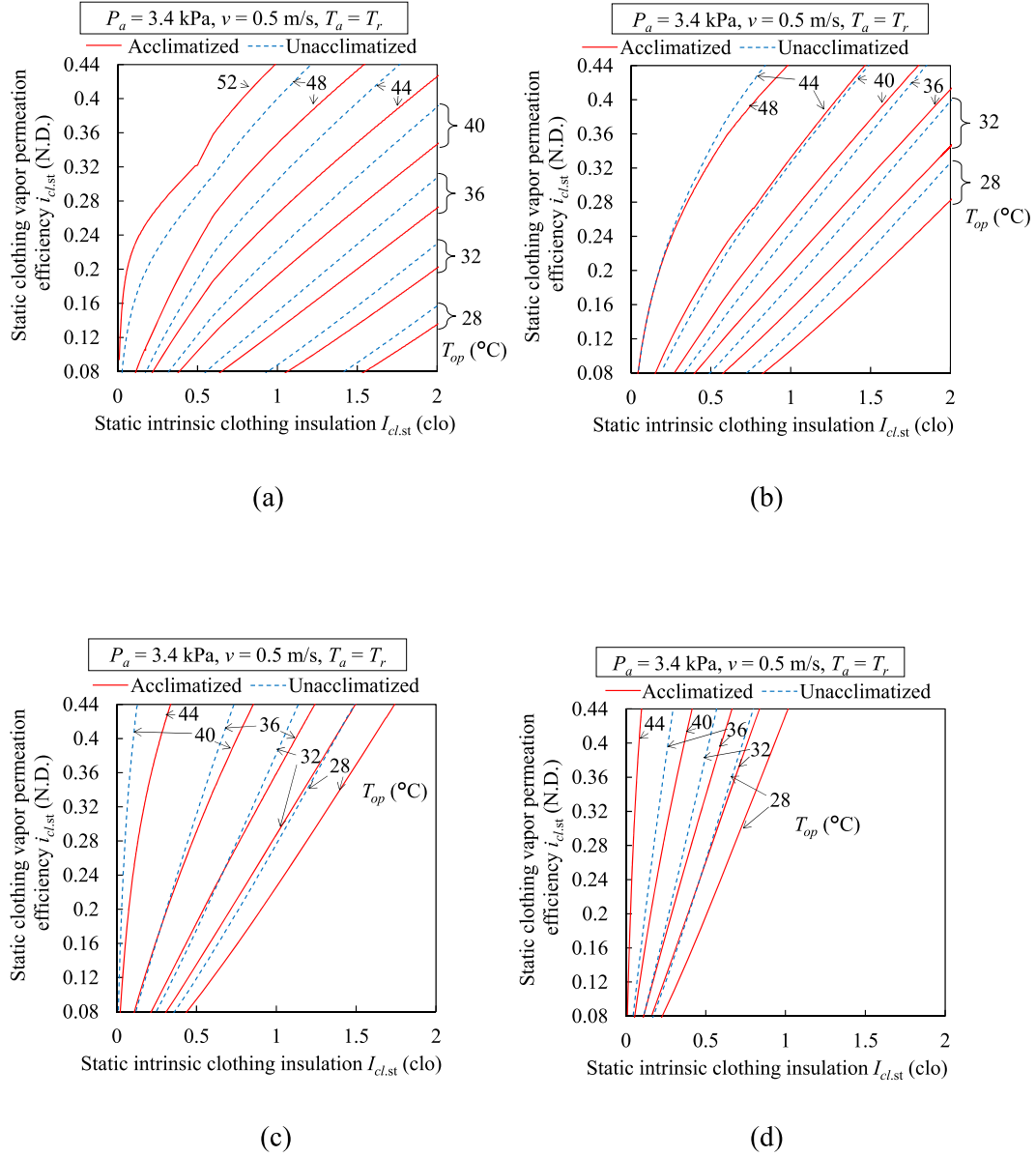
The thresholds are drawn as curves connecting the points of low  $I_{cl,st}$  and  $i_{cl,st}$  to the points of high  $I_{cl,st}$  and  $i_{cl,st}$ . The curves tend to converge at a point beyond the bottom left corner of Fig. 6. The static evaporative heat transfer resistance of clothing ( $R_{e,cl,st}$ ) is related to  $I_{cl,st}$  and  $i_{cl,st}$ , as shown in Eq. (12) [44]; an increase in  $i_{cl,st}$  for a fixed  $I_{cl,st}$  lowers  $R_{e,cl,st}$ :

$$R_{e,cl,st} = \frac{0.155 I_{cl,st}}{LR i_{cl,st}}, \quad (12)$$

where  $R_{e,cl,st}$  is a static intrinsic clothing evaporative resistance (m<sup>2</sup>kPa/W).

By considering the relationships in Eq. (12), we conclude that the constant  $R_{e,cl,st}$  lines in Fig. 6 must be straight lines converging at a point beyond the bottom left corner of Fig. 6, where both  $I_{cl,st}$  and  $i_{cl,st}$  are zero.





**Fig. 6.**  $I_{cl,st}$  and  $i_{cl,st}$  resulting in the same threshold  $T_{op}$  at various activity levels: (a)  $M$  at  $58 \text{ W/m}^2$ , (b)  $M$  at  $116 \text{ W/m}^2$ , (c)  $M$  at  $274 \text{ W/m}^2$ , and (d)  $M$  at  $233 \text{ W/m}^2$ .

The similarity between the  $I_{cl,st}$  and  $i_{cl,st}$  combinations that resulted in the same threshold  $T_{op}$  and those that resulted in the same  $R_{e,cl,st}$  suggests that  $R_{e,cl,st}$  is the primary determinant of the threshold when both  $I_{cl,st}$  and  $i_{cl,st}$  vary. Regarding the effect of the work intensity (metabolic rate), the threshold lines at a constant  $T_{op}$  become denser with an increase in  $M$ . This finding means that the influence of  $I_{cl,st}$  and  $i_{cl,st}$  on the threshold increases with the work intensity.

#### 4. Discussion

The thresholds determined by the heatstroke threshold equation are  $T_{cr}$  ( $T_{cr,lim}$ ) of  $38.0 \text{ }^{\circ}\text{C}$  for unacclimatized workers and  $38.5 \text{ }^{\circ}\text{C}$  for acclimatized workers. In contrast, the upper limits for the prescriptive zone were defined as the point at which an inflection in  $T_{cr}$  was observed for the change in the surrounding environment in human trials. Therefore, the threshold determined by the heatstroke threshold equation does not have the same meaning as the upper limit of the prescriptive zone in terms of  $T_{cr}$ . However, because both thresholds and upper limits provide the boundaries for workers engaged in continuous work, we present a comparative analysis of the thresholds and upper limits in this

section.

Bernard et al. [13,45] reported the upper limit of the prescriptive zone for acclimatized workers wearing one of five clothing ensembles, based on studies of human subjects walking on a treadmill. The types of clothing ensembles considered in their studies were work clothes (WC), cotton coveralls (CC), and Tyvek, NexGen, and Tychem QC (TYC). The reported  $M$  values ranged from  $115$  to  $254 \text{ W/m}^2$ . The humidity was set to three levels of 20, 50, and 70 % RH for  $M = 160 \text{ W/m}^2$ , whereas it was set to a single level of 50 % RH for the conditions of  $M = 115, 180$ , and  $254 \text{ W/m}^2$ . We validated the heatstroke threshold equation by comparing the thresholds with the reported data [13,45]. The total thermal insulation and total evaporative resistance values, measured using a thermal manikin, were provided for WC, CC, and TYC by Prof. Bernard through personal communication. These values were converted into the intrinsic clothing insulation ( $I_{cl}$ ) and clothing vapor permeation efficiency ( $i_{cl}$ ) and applied to the heatstroke threshold equation. Then, we determined the thresholds for WC, CC, and TYC for acclimatized workers. Bernard et al. reported the upper limits of the prescriptive zone based on WBGT at 50 % RH for  $M = 115, 180$ , and  $254 \text{ W/m}^2$ . Therefore, we estimated  $T_{op}$  at 50 % RH, corresponding to the reported upper-limit

**Table 1**

Calculation conditions reported in this section.

| Parameters                                                      | Fig. 4                                                                                                                                                                                                                                                                            | Fig. 5                                                    | Fig. 6                                                                                                              |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Variables focused on to consider mutual influence on thresholds | Operative temp. $T_{op}$ (°C) and water vapor pressure $P_a$ (kPa).                                                                                                                                                                                                               | Operative temp. $T_{op}$ (°C) and air velocity $v$ (m/s). | Static clothing insulation $I_{cl, st}$ (clo) and static clothing vapor permeation efficiency $i_{cl, st}$ (N. D.). |
| Clothing                                                        | (a) Work clothes (WC): $I_{cl, st} = 0.6$ , $i_{cl, st} = 0.34$ ,<br>(b) Vapor-permeable but air-impermeable protective clothing (VPAIP): $I_{cl, st} = 0.6$ , $i_{cl, st} = 0.34$ ,<br>(c) Film-laminated protective clothing (FLP): $I_{cl, st} = 1.13$ , $i_{cl, st} = 0.09$ . | –                                                         | –                                                                                                                   |
| $P_a$ (kPa)                                                     | –                                                                                                                                                                                                                                                                                 | 3.4 (60 % RH at 35 °C)                                    | –                                                                                                                   |
| $v$ (m/s)                                                       | 0.5                                                                                                                                                                                                                                                                               | –                                                         | 0.5                                                                                                                 |
| Metabolic rate $M$ (W/m <sup>2</sup> )                          | 58, 116, 174, 233                                                                                                                                                                                                                                                                 | –                                                         | –                                                                                                                   |
| Acclimatization level                                           | Unacclimatized, acclimatized.                                                                                                                                                                                                                                                     | –                                                         | –                                                                                                                   |
| $T_{op}$ (°C)                                                   | –                                                                                                                                                                                                                                                                                 | –                                                         | ≥28 °C at 4 °C intervals                                                                                            |
| Others                                                          | Isothermal environment, $\epsilon_{cl} = 0.95$ , $F_r = 0.8$ , $h_c$ for standing posture of de Dear et al. [41], Influence of activity and air velocity is taken into account referring to Refs. [42,43].                                                                        |                                                           |                                                                                                                     |

WBGT, by solving the heat balance equations of the natural wet-bulb and black globe thermometers. Fig. 7 compares  $T_{op}$  and  $P_a$  of the thresholds and those of the reported upper-limit environments for the prescriptive zone [13,45].

The threshold coincides with the reported upper limit for the prescriptive zone within 0.5 °C of  $T_{op}$  at the same  $P_a$  in an environment of approximately 20 % RH with  $M$  of 160 W/m<sup>2</sup> for TYC. In all other cases, the thresholds were located at a lower  $T_{op}$  level than the reported upper limits for the prescriptive zone. Because thresholds at a lower  $T_{op}$  than the reported level can be considered to contribute to risk reduction, the heatstroke threshold equation can provide the critical reference value to prevent the occurrence of heatstroke for various combinations of  $T_{op}$ , RH, and  $M$ .

Regarding the acclimatization levels and air velocity differences, Kenny et al. [46] compared the upper limits of the prescriptive zone for female subjects. We validated the heatstroke threshold equation by comparing the thresholds with previously reported data [46]. Fig. 8 compares the thresholds with the reported upper limits for the prescriptive zone. For condition 2, we adopted an air velocity of 0.15 m/s instead of the reported value of 0 m/s because air was supplied to the chamber in the experiments by Kenny et al. [46].

When the subjects were unacclimatized workers (Cond. 1), the threshold was located at a higher  $T_{op}$  level than the reported upper limit for the prescriptive zone. The heatstroke threshold equation underestimated the risk of heatstroke in this case. It is desirable to reconsider the parameters set in our heat balance equation for unacclimatized workers to improve the evaluation accuracy.

The thresholds of the heatstroke threshold equation coincided with the reported upper limits for the prescriptive zone for acclimatized workers (Cond. 2 and Cond. 3, which differ in terms of the air velocity). Therefore, the heatstroke threshold equation can be used to evaluate the effect of air velocity on acclimatized workers.

Bernard et al. indicated the relationship between the static clothing total evaporative resistance ( $R_{e, T, st}$ ) and WBGT clothing adjustment value (CAV) based on the upper limit of the prescriptive zone [47]. Adding the CAV to the measured WBGT in the evaluated condition converts the WBGT into a WBGT effective scale for workers wearing the summer work uniform. We validated the heatstroke threshold equation by comparing the CAV determined from the equation with the reported CAV corresponding to the upper limit of the prescriptive zone. To

determine the CAV from the equation, we firstly estimated the threshold  $T_{op}$  for several combinations of the static intrinsic clothing evaporative resistance ( $R_{e, cl, st}$ ) and static intrinsic clothing thermal insulation ( $I_{cl, st}$ ) under isothermal conditions with 50 % RH,  $v$  of 0.5 m/s, and  $M$  of 160 W/m<sup>2</sup> for acclimatized workers. Then, we determined the natural wet-bulb temperature ( $T_{nw}$ ) and black globe temperature ( $T_g$ ) by solving the heat balance equations of the natural wet-bulb and black globe thermometers and calculated the WBGT for each threshold  $T_{op}$  at 50 % RH. We determined the CAV by subtracting the WBGT calculated for the combination of  $R_{e, cl, st}$  and  $I_{cl, st}$  from the WBGT calculated for workers wearing summer work uniforms. Fig. 9 shows the comparison of the calculated CAVs with the reported CAVs. Before drawing Fig. 9, we converted  $R_{e, cl, st}$  used in the calculation into  $R_{e, T, st}$  by adding the air-layer resistance under calm air conditions, which was corrected beforehand using the clothing area factor.

The determined CAV is much larger than the reported CAV when  $R_{e, T, st}$  is exceptionally high. This high  $R_{e, T, st}$  value corresponds to that of a capsule suit with a negative-pressure respirator and is the highest among the values of the measured protective clothing ensembles in the Level B classification. Except for this datum, the CAVs determined using the heatstroke threshold equation agree with the reported CAVs. This finding means that the heatstroke threshold equation can be used to evaluate the quantitative influence of clothing, except for capsule suits with an excessively high  $R_{e, T, st}$ .  $R_{e, T, st}$  is the sum of the clothing static intrinsic evaporative heat transfer resistance ( $R_{e, cl, st}$ ) and air-layer evaporative resistance under calm air conditions, corrected with the clothing area factor. Because the change in  $R_{e, cl, st}$  is much larger than the change in the air-layer evaporative resistance in calm air corrected with the clothing area factor,  $R_{e, cl, st}$  is considered the dominant factor for  $R_{e, T, st}$ . Considering that all the CAV lines of  $I_{cl, st}$  converge on the left side of Fig. 9, we can conclude that  $R_{e, cl, st}$  is the primary determinant factor related to clothing for the thresholds when  $R_{e, cl, st}$  and  $R_{e, T, st}$  are small.

In this section, we present the heatstroke threshold equation based on our series of studies. Sakoi and Mochida proposed an equivalent wet-bulb globe temperature index (eWBGT) [43]. The eWBGT expresses thermal environments in the scale of the WBGT for workers wearing summer work uniforms based on the heat balance equation of the human body. Sakoi et al. expressed the human heat balance equation as a WBGT equation and clarified how the coefficients of the natural wet-bulb temperature, black globe temperature, and air temperature change depending on clothing and air velocity [48]. Sakoi et al. further expanded the previously derived expression to a thermal environment index, namely, the new effective wet-bulb globe temperature (WBGT<sub>eff</sub>\*). WBGT<sub>eff</sub>\* can express thermal environments using the WBGT effective scale for workers wearing a summer work uniform by changing the coefficients of the natural wet-bulb temperature, black globe temperature, and air temperature, and the intercept depending on the type of clothing and air velocity [49]. The human heat balance equation, which is adopted in eWBGT and WBGT<sub>eff</sub>\*, focused on the skin, clothing, and surrounding environment. However, the heat balance in the core body is not considered. The heatstroke threshold equation formulated in this study includes the core body temperature in the human body heat balance because the rise in core body temperature provides the limit for heatstroke risk management, regardless of the activity [5–8]. By defining the upper-limit core temperature ( $T_{cr, lim}$ ), thermal resistance between the body core and skin with the maximum blood flow ( $R_{cs, min}$ ), and the maximum sweat rate ( $SW_{max}$ ) at a given intensity of activity, the heatstroke equation enables the determination of the upper limit for occupational health. The equation can be applied to workers performing various tasks with different types of clothing and working in a thermal environment with a combination of various air temperatures, humidity, radiation, and air velocity. However, because the proposed method was validated using  $M$  less than or equal to 254 W/m<sup>2</sup>, the heatstroke threshold equation can be applied to situations with  $M$  lower than 254 W/m<sup>2</sup>.

Further, the heatstroke threshold equation cannot be used to

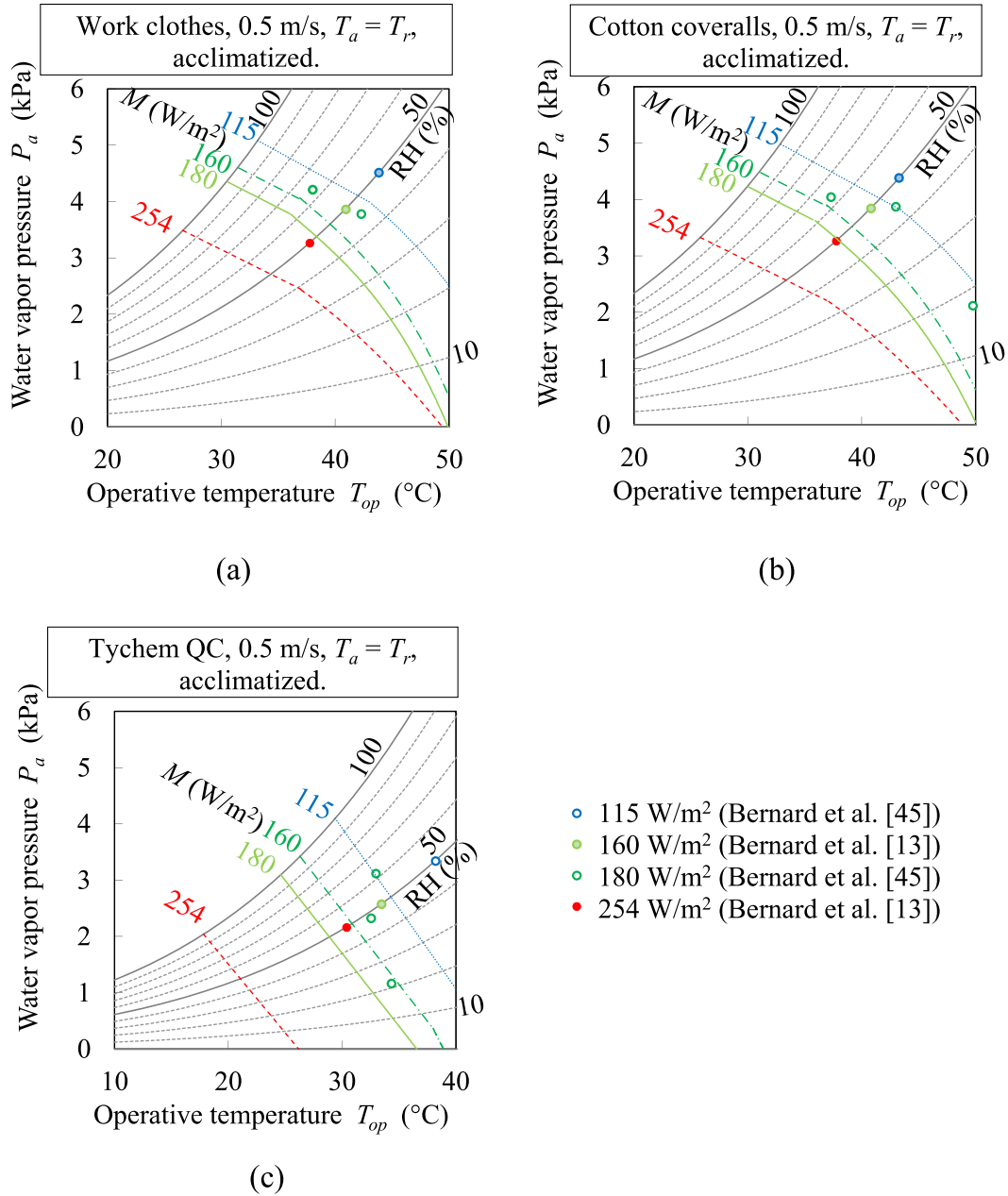
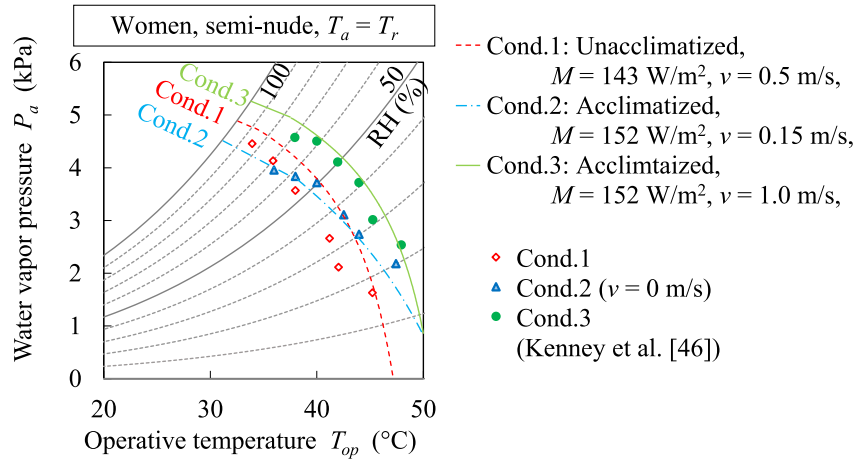


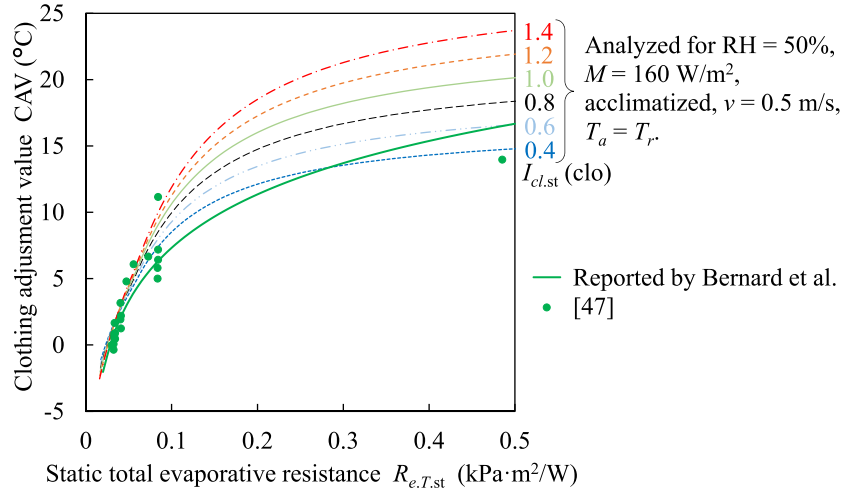
Fig. 7. Thresholds based on our heatstroke threshold equation and reported upper limits for the prescriptive zone: (a) WC, (b) CC, and (c) TYC.

evaluate the transient response. It can only assess whether the thermal condition exceeds the acceptable limit. Therefore, the heatstroke threshold equation has limited applicability. It is similar in concept to the comfort equation [50], which includes six main factors for calculating the thermal comfort status: air temperature, mean radiant temperature, humidity, air velocity, clothing, and metabolic rate. A comfortable status satisfies the equilibrium between the human body heat generation and heat dissipation from the body with a comfortable skin temperature and sweat rate. Because the core body temperature is critical for evaluating the risk of heatstroke,  $T_{cr,lim}$  was first defined for the heatstroke threshold equation. Then, we solved the steady heat transfers between the body core, skin, clothing, and surrounding environment, as illustrated in Fig. 2. We compared the heat generation at the body core with the steady heat dissipation from it to evaluate whether heat accumulation occurred in a body core with  $T_{cr,lim}$ . Although the simple evaluation limits the applicability of the heatstroke threshold equation, it reduces the required parameters and formulas. The

parameters and formulas required for the human body are  $T_{cr,lim}$ ,  $R_{cs,min}$ ,  $SW_{max}$ , and the sweat evaporation efficiency ( $\eta$ ). The other required parameters and formulas are the heat transfer characteristics between the skin, clothing, and surrounding environment, as well as respiration heat exchange formulas. The threshold for the core body temperature exceeding the acceptable limit is in line with the reported limit of the prescriptive zone for acclimatized workers in conditions with various combinations of activities, clothing, airflow, operative temperature, and humidity. Therefore, we conclude that the parameters and formulas adopted in the heatstroke threshold equation are determinants of the critical boundary value for heatstroke occurrence. However, as previously reported, the heatstroke threshold equation overestimates the thresholds and underestimates the risk for unacclimatized workers. The parameters and formulas used in this study were set by referring to average data of unacclimatized workers or acclimatized workers reported in the existing papers. As a limitation of the heatstroke threshold equation, we did not conduct measurement designed for setting



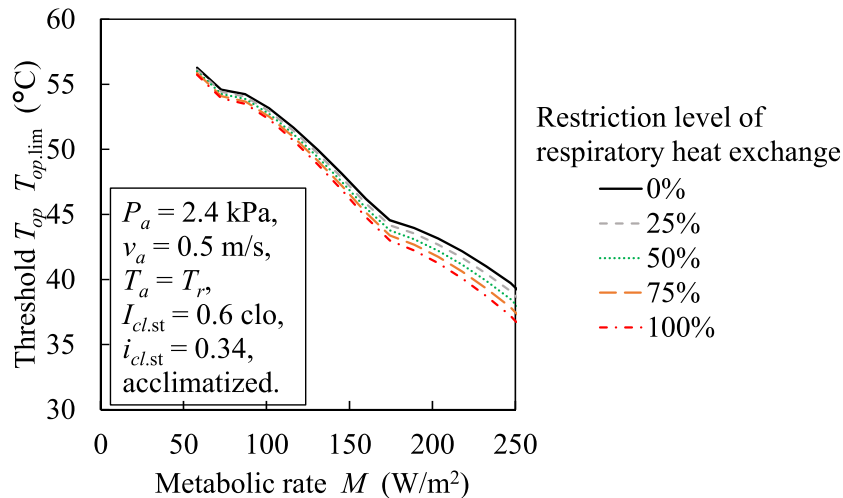
**Fig. 8.** Thresholds from our heatstroke threshold equation and reported upper limits for the prescriptive zone for different air velocities and heat acclimatization levels.



**Fig. 9.** Comparison of CAVs determined from our heatstroke threshold equation with reported CAVs for  $R_{e.T.st}$  and  $I_{cl.st}$  combinations.

parameters and formulas of the equation. Therefore, the accuracy improvement can be expected by replacing the parameters and formulas based on measurement designed for setting them. To evaluate the

heatstroke risk in individual cases, including unacclimatized workers, parameter and formula updates are required to improve the accuracy. In addition, in this study, we applied and validated the equation based on



**Fig. 10.** Threshold  $T_{op}$  for different restriction levels of respiratory heat exchange.

studies on subjects walking on a treadmill. However, occupational environments involve various types of activities. Therefore, setting the heat transfer characteristics between the skin, clothing, and surrounding environment by considering different activities and individual differences is preferred. Although the body size and shape investigated in our previous studies [51–53] were those of an average infant and differed widely from those of occupational workers, we reported the procedures for determining the convective heat transfer coefficients [51], clothing insulation [52], and clothing area factors [53] for the human body.

Finally, we provide an example of the application of the heatstroke threshold equation. We evaluated the effect of masks on the heatstroke risk. Fig. 10 shows the threshold  $T_{op}$  ( $T_{op,lim}$ ) at several restricted levels of respiratory heat exchange for various activities under the conditions of  $P_a = 2.4$  kPa,  $v_a = 0.5$  m/s, an isothermal environment, and summer work uniforms being worn by the workers. We treated the influence of the mask as a restriction on the respiratory heat exchange. Fig. 10 shows the results for acclimatized workers.

When  $M$  is 58.15 W/m<sup>2</sup>, the difference in  $T_{op,lim}$  is 0.5 °C, whereas it is 4.5 °C when  $M$  is 250 W/m<sup>2</sup>. When the metabolic rate is low, the influence of the mask is small. However, the mask restriction appears more significant when the metabolic rate is high. The mask increases the risk of heatstroke because the respiratory heat exchange is restricted to a greater extent under an increased metabolic rate.

## 5. Conclusions

Skin temperature and sweat rate are critical for thermal comfort, whereas the core body temperature is critical for assessing the risk of heatstroke. To realize an integrated design of clothing, airflow around the human body, and indoor and outdoor environments as a single system, we illustrated the heat interactions between the core body, skin, clothing, airflow around the human body, and indoor and outdoor environments as water flows caused by the water head difference. In the illustration, the water flow, water head, and cross-sectional area of the water tank express the heat flow, temperature, and heat capacity, respectively.

Based on the illustrated model, we propose a heatstroke threshold equation to evaluate whether a given combination of six factors—air temperature, mean radiant temperature, humidity, airflow, clothing, and metabolic rate—causes heatstroke. The heatstroke threshold equation includes the core body heat balance and heat balance between the skin, clothing, and surrounding environment. It expresses the heatstroke risk by comparing the heat generation in the core body with the steady heat dissipation from it to determine if heat accumulation occurs in the core body with the acceptable limit temperature. Then, it provides the threshold values for the combination of six factors for the body core temperature to exceed the acceptable limit. The threshold changes depending on the air temperature, mean radiant temperature, humidity, airflow, clothing, and metabolic rate.

The heatstroke threshold equation was validated based on studies that have reported the upper limits of the prescriptive zone. The heatstroke threshold equation successively determined the thresholds for acclimatized workers whereas it underestimated the heatstroke risk for unacclimatized workers. Nevertheless, the heatstroke threshold equation can be an effective tool for the risk management of acclimatized workers for ensuring occupational health because it provides a rational heatstroke risk boundary for workers engaging continuously in activities of various intensities, wearing various clothing ensembles, and working in thermal environments with various combinations of air temperature, humidity, thermal radiation, and airflow.

Shortages of water and salt in the body or poor health of the worker can lead to heat disorders such as cramps and exhaustion, even in an environment below the threshold. Because such initial heat disorders enhance the occurrence of heatstroke, the heatstroke threshold equation should be applied only to workers who are in good health and have sufficient water and salt intake. In addition, this method underestimated

the heatstroke risk in unacclimatized workers. Future research should focus on updating the parameters and formulas used in this equation. In addition, because the validated range of the metabolic rate for the heatstroke threshold equation was limited to 254 W/m<sup>2</sup> or less, the equation is applicable to scenarios where the metabolic rate is within this range.

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## CRediT authorship contribution statement

**Tomonori Sakoi:** Writing – review & editing, Writing – original draft, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yoshihito Kurazumi:** Conceptualization. **Sri Rahma Apriliyanthi:** Writing – review & editing, Data curation. **Shin-ichi Sawada:** Resources, Methodology. **Chuansi Gao:** Writing – review & editing, Methodology.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Tomonori Sakoi reports financial support was provided by Japan Science and Technology Agency. Tomonori Sakoi reports financial support was provided by Japan Society for the Promotion of Science. Tomonori Sakoi reports financial support was provided by TORAY Industries, Inc.

## Data availability

The authors do not have permission to share data.

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